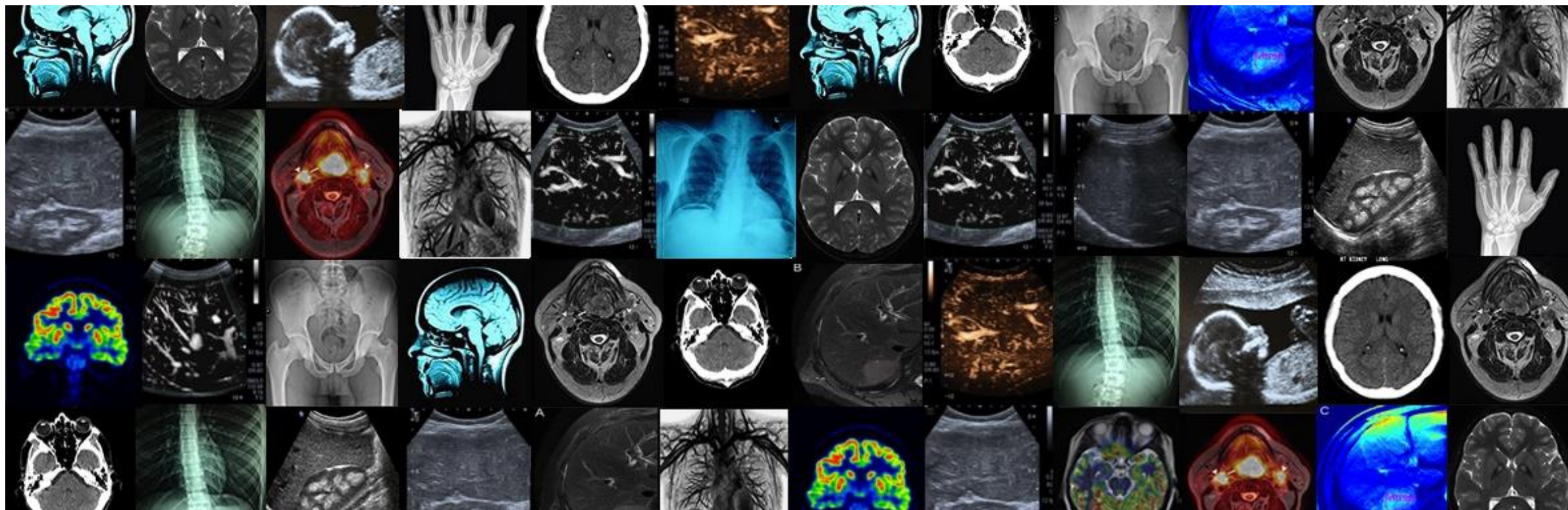




Summer Student program 2026 **Enhanced imaging data format support for AI in PET**

Author: Andrea Giovannelli

Supervisors: Jens Maus, Pavel Nikulin

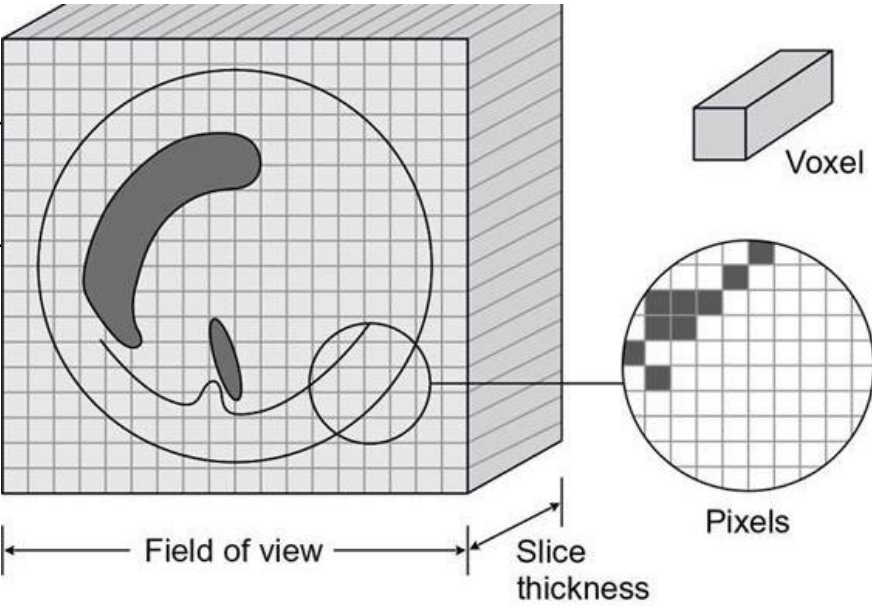
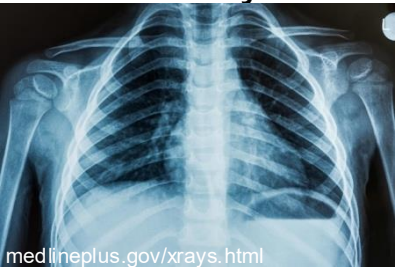
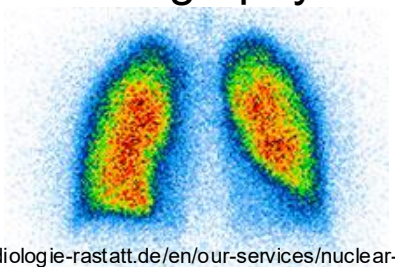
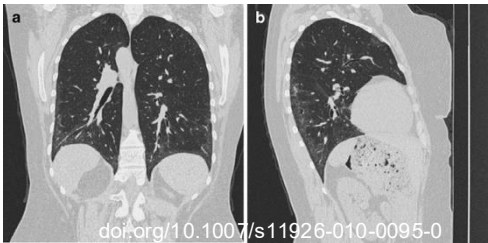
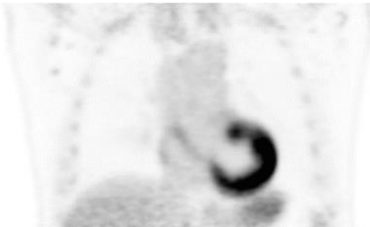
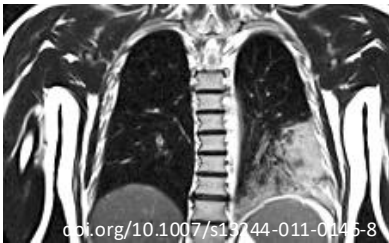
DRESDEN
concept



INTRODUCTION

Medical Imaging Techniques Tomographic modalities

Medical Imaging is a common name for a set of techniques allowing to obtain images of the human anatomy and physiology via minimally invasive methods.

	Ionizing radiation		
	Transmission	Emission	
Planar (2D)	X-Ray  medlineplus.gov/xrays.html	Scintigraphy  radiologie-rastatt.de/en/our-services/nuclear-medicine/lung/	
Tomographic(3D)	Computed Tomography (CT)  doi.org/10.1007/s11926-010-0095-0	Positron Emission Tomography (PET) and SPECT¹  ¹ Single Photon Emission Computed Tomography	Magnetic Resonance Imaging (MRI)  doi.org/10.1007/s11944-011-0145-8

Picture adapted from [Computed Tomography](#)

DATA FORMATS

ECAT

- Legacy PET format, developed for Siemens/CT scanners;
- Stores Header (acquisition/physics metadata) + Image Data;
- Contains headers, image dimensions, timing & calibration factors, physical metadata (half-life, decay, dosage) necessary for quantitative PET analysis.
- **Limitations:** fixed structure; limited spatial information; poor support in modern AI workflows

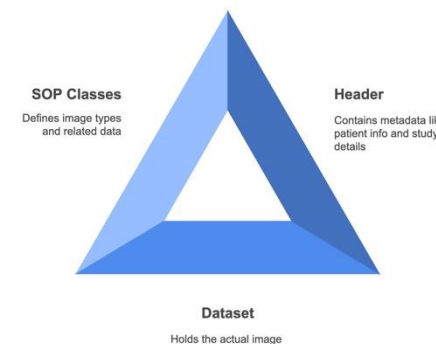
Analyze

- Legacy binary format for PET & MRI;
 - Limited and fixed-size header;
 - **Limitations:** poorly defined patient orientation.
- NIfTI preferred for modern analysis.

DICOM

Digital Imaging and Communications in Medicine

- The universal clinical standard for the storage and management of medical imaging information;
- It is used for CT, MRI, PET, X-ray, ultrasound and mammography;
- A DICOM file contains: Header (metadata) + Voxel Matrix (Image) + SOP Classes;
- **Limitations:** complex and fragmented; not an appropriate format for data processing or AI workflows.



NIfTI

Neuroimaging Informatics Technology Initiative

- The standard for neuroimaging and AI;
- Fast and monolithic;
- **Limitations:** lacks native header fields to store critical PET physical metadata.

Many others: **Interfile**, **HDF5**, **MINC**, **NRRD**, **MetaImage** ...

Goal: Bridge these formats using libmedio (MedIOData)

MOTIVATION

NIfTI, **Interfile** and **HDF5** are widely used in research and data-processing workflows, including modern AI-based imaging pipelines. However, our own main PET analyses tool (**ROVER - ROI Visualization, Evaluation and Image Registration**) does not directly support these formats yet.

Currently ECAT is the main PET data format used in ROVER, but should be potentially enhanced with Nifti, Interfile or hdf5 in future.

Project Main goals:

1. Implement NIfTI and INTERFILE formats into libmedIO (<https://github.com/hzdr-MedImaging/libmedio>);
2. Write test cases to verify correct **read/write support** for nifti and interfile;
3. Write data conversion tools "**ecat2nifti**", "**ecat2interfile**", etc. which are using libmedio;
4. ROVER tools should directly support NIfTI → not part of the summer student project ...

NIfTI-1 & NIfTI2 File

File Architecture

Monolithic File Structure (.nii):

Stored as a continuous binary stream combining metadata headers and data.

1. Main Header (The "Instruction Manual")

- NIfTI-1: Fixed size of 348 bytes (Bytes 0–347), utilizing standard 16/32-bit fields for dimensions and offsets.
- NIfTI-2: Expanded size of 540 bytes, designed with 64-bit.

2. Magic Number Signatures (Format Validation):

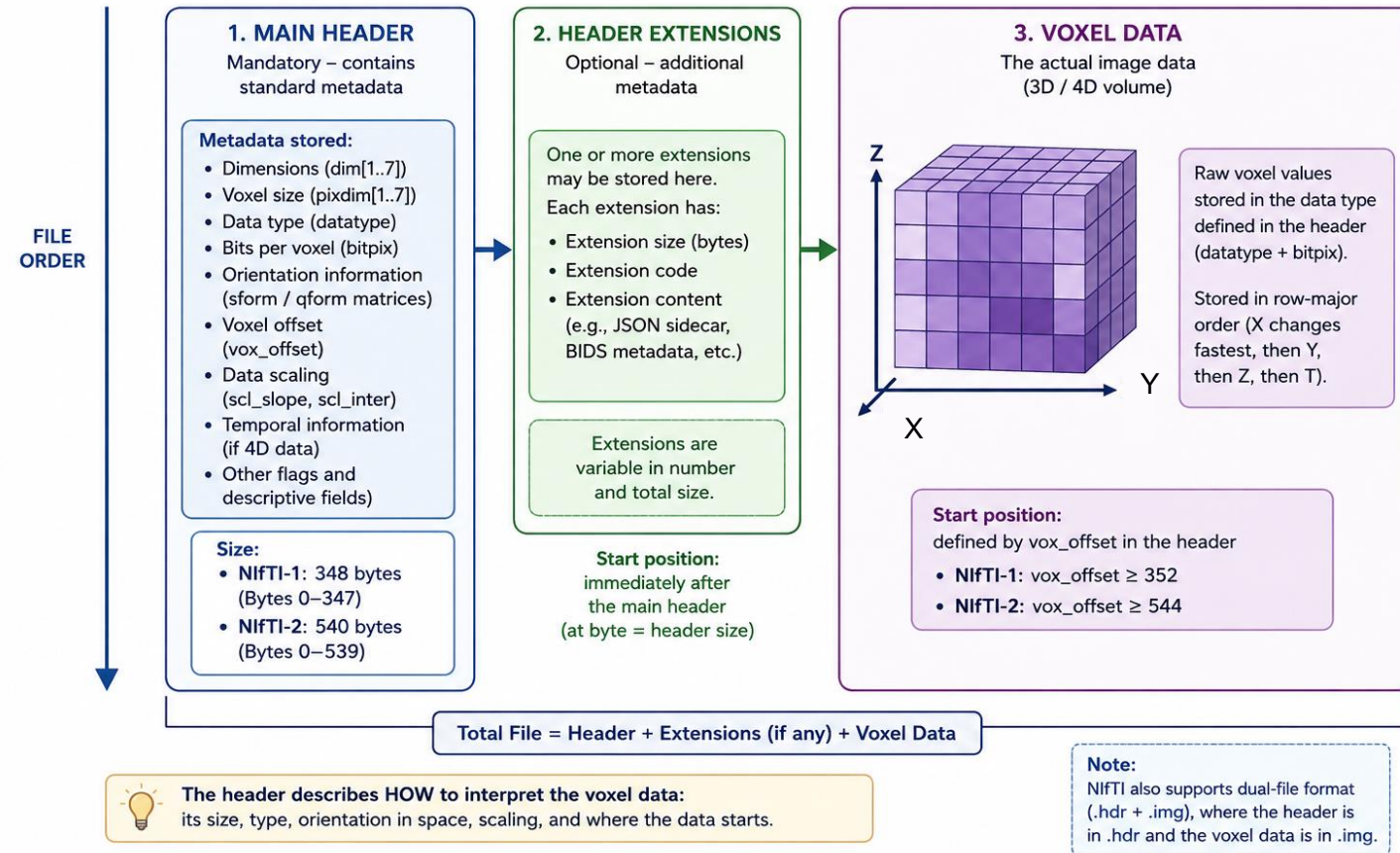
- NIfTI-1: Uses `n+1\0` (single file) or `ni1\0` (split files) located at bytes 344–347.
- NIfTI-2: Uses `n+2\0` or `ni2\0` located at the end of the larger 540-byte header.

3. Header Extensions & Voxel Offset: Accommodates optional metadata extensions (such as BIDS JSON, size of 4 bytes). Vox_Offset points to where pixel data begins (≥ 352 for NIfTI-1, ≥ 544 for NIfTI-2).

4. Voxel Matrix Data: The actual 3D/4D image (radioactivity concentrations / clinical intensities).

NIFTI FILE ARCHITECTURE

Typical .nii / .nii.gz file (single-file format)



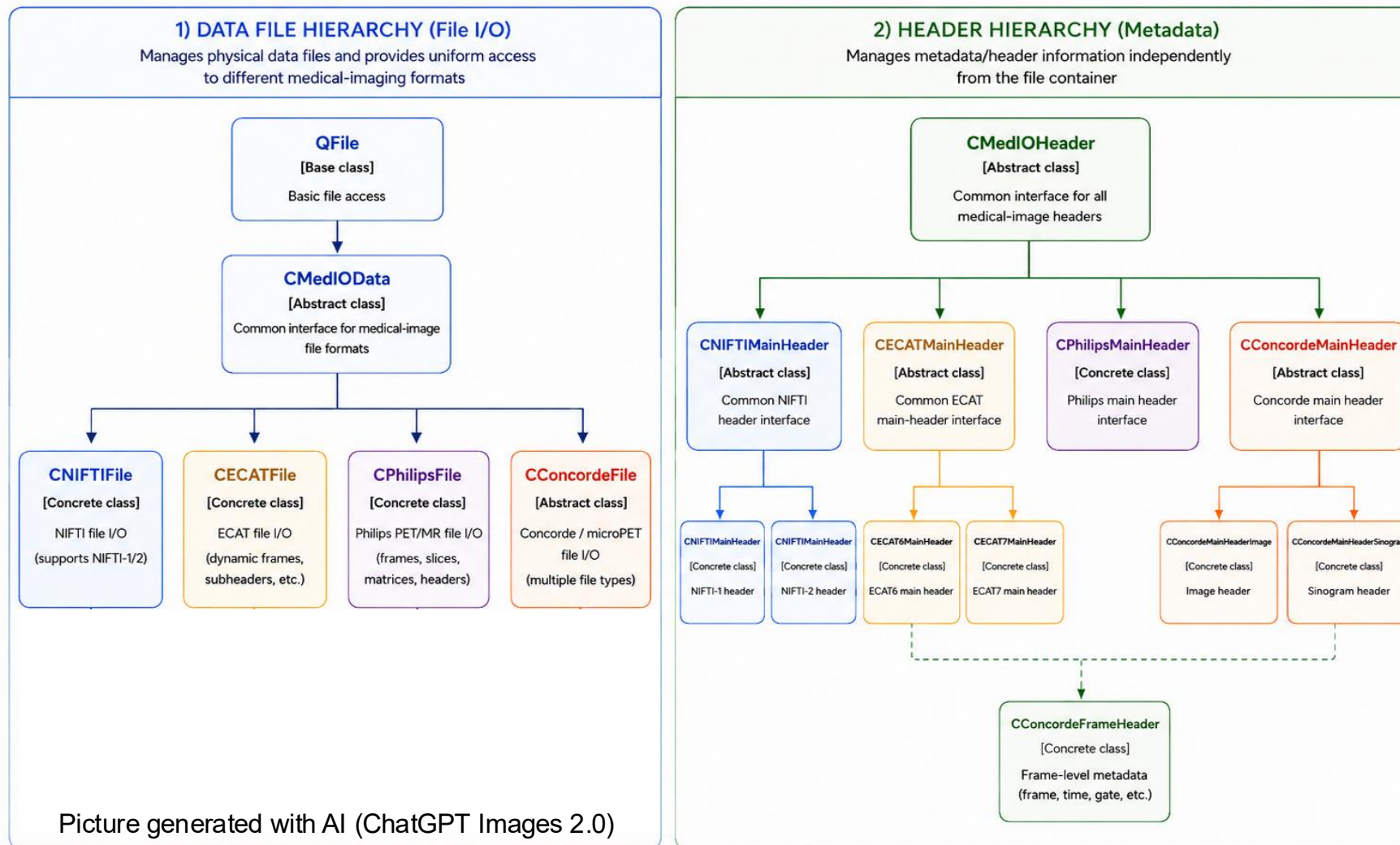
Picture generated with AI (ChatGPT Images 2.0)

libmedio - medical imaging I/O library

Scheme of C++/Qt based library

libmedio is a medical-imaging I/O library, structured to make a clear distinction between the management of data files and of header metadata.

libmedio C++ Architecture — File I/O and Metadata Class Hierarchy



Picture generated with AI (ChatGPT Images 2.0)

Spatial Alignment

Qform & Sform transformation

Following the **ECAT** ↔ **NIfTI** conversion:

- Z-axis is inverted: the order in which the slices are represented is reversed;
- X & Y axes are flipped.

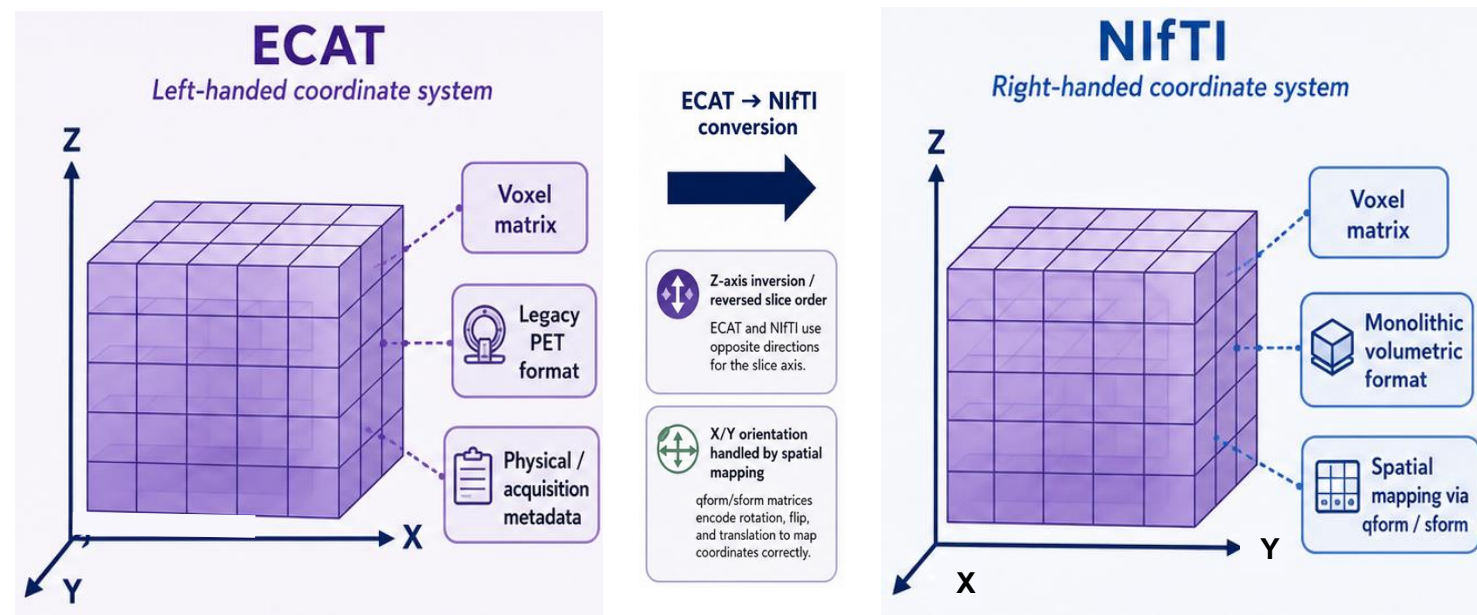
Qform (Scanner-Anatomical)

It maps voxel indices (i,j,k) to scanner-based anatomical coordinates (x,y,z) using Quaternion representation [a, b, c, d] (to store a rotation with only 4 parameters instead of 9). Implemented in order to handle scaling and translation offsets derived from the ECAT bed position. Pixdim and offset are stored in metadata.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = R \cdot \begin{pmatrix} \text{pix_dim}[1] \cdot i \\ \text{pix_dim}[2] \cdot j \\ qfac \cdot \text{pix_dim}[3] \cdot k \end{pmatrix} + \begin{pmatrix} \text{qoffset_x} \\ \text{qoffset_y} \\ \text{qoffset_z} \end{pmatrix}$$

where

$$R = \begin{pmatrix} a^2 + b^2 - c^2 - d^2 & 2bc - 2ad & 2bd + 2ac \\ 2bc + 2ad & a^2 + c^2 - b^2 - d^2 & 2cd - 2ab \\ 2bd - 2ac & 2cd + 2ab & a^2 + d^2 - b^2 - c^2 \end{pmatrix}$$



Picture generated with AI (ChatGPT Images 2.0)

Sform (Affine Transformation)

Defines a general affine transformation to standard space. Set sform_code = 1 to map the left-handed ECAT data into a right-handed NIfTI.

$$\begin{cases} x = \text{Srow_X}[0] \cdot i + \text{Srow_X}[1] \cdot j + \text{Srow_X}[2] \cdot k + \text{Srow_X}[3] \\ y = \text{Srow_Y}[0] \cdot i + \text{Srow_Y}[1] \cdot j + \text{Srow_Y}[2] \cdot k + \text{Srow_Y}[3] \\ z = \text{Srow_Z}[0] \cdot i + \text{Srow_Z}[1] \cdot j + \text{Srow_Z}[2] \cdot k + \text{Srow_Z}[3] \end{cases}$$

libmedio NIfTI-1 & NIfTI-2 Implementation Summary

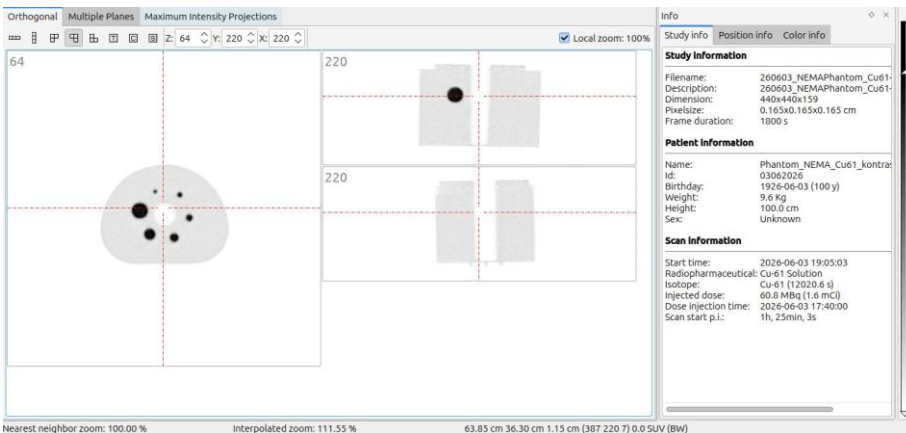
- **NIfTI-1 and NIfTI-2 Support:** Implemented read and write capabilities for both NIfTI-1 and NIfTI-2 formats within libmedio.
- **BIDS (Brain Imaging Data Structure) JSON Metadata Integration:** Implemented JSON file generation to store PET acquisition metadata (isotopes, half-life, ect.) and 4D dynamic parameters. This step was crucial because such information would otherwise be lost during the conversion process from ECAT to NIfTI;
- **ECAT-to-NIfTI-1/NIfTI-2 bidirectional Conversion (`ecat2nifti` & `nifti2ecat`) & Spatial Alignment:** Developed conversion modules between ECAT and NIfTI formats. Geometric alignment to ensure that original and converted images correspond accurately. This was achieved by applying proper spatial transformations using `qform_code` and `sform_code`;
- **Example Applications & Utilities:** Added dedicated test examples and verification utilities (such as `showheader_nifti` and updates to existing ECAT example modules) to validate the new input/output and conversion pipelines;
- **NIfTI Header Extensions:** Initiated the implementation of NIfTI header extensions (along with BIDS header extension support).

Round-Trip Validation

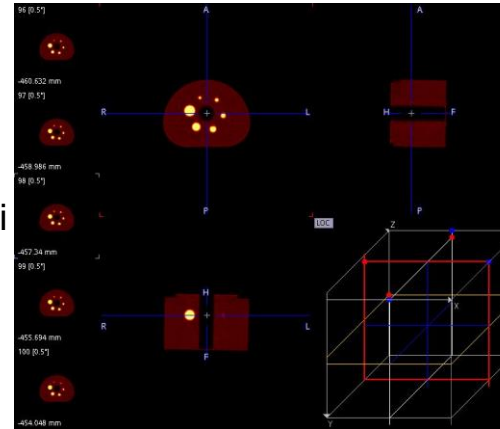
PMOD & ROVER displayer

Visual validation of the ECAT↔NIfTI conversion pipeline using a NEMA IEC PET phantom scan (*Cu-61*, List-Mode, PSF+TOF reconstruction), visualized in the PMOD medical imaging suite.

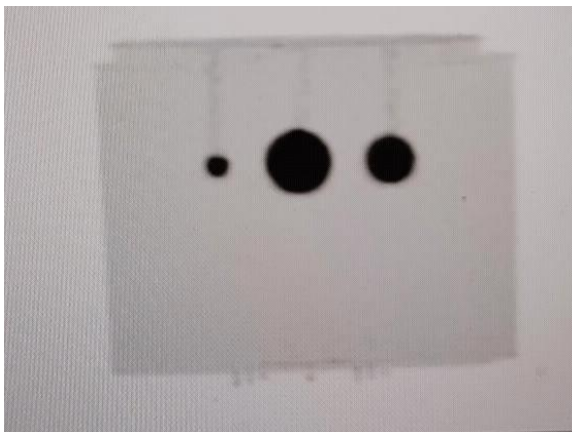
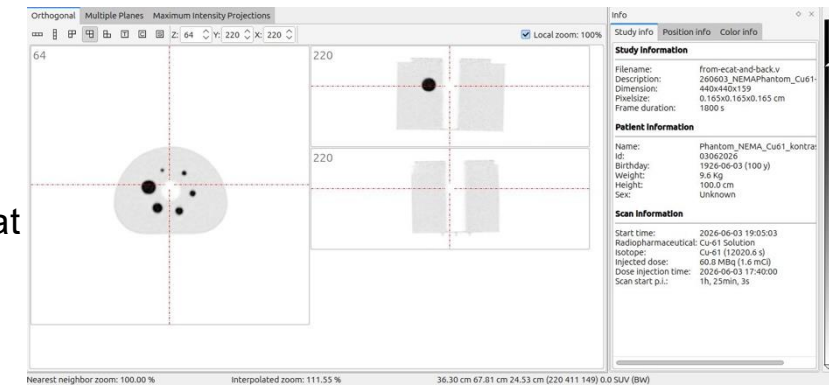
Testing **ECAT** → **NIfTI** → **ECAT** round-trip



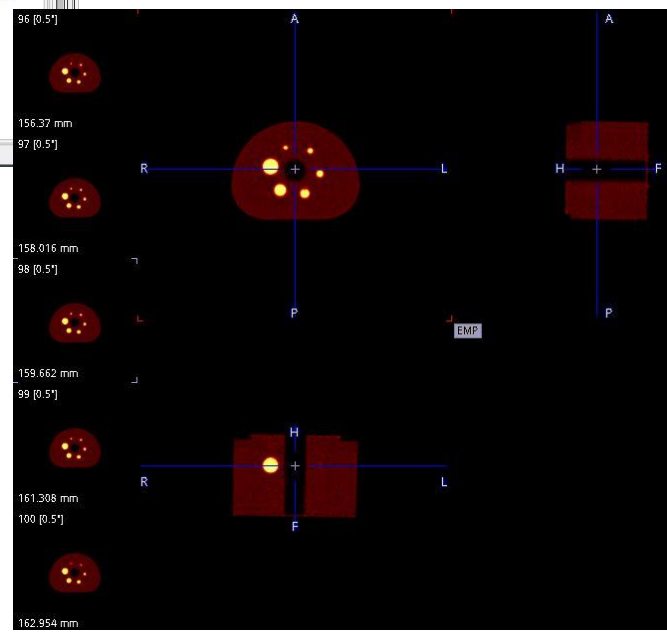
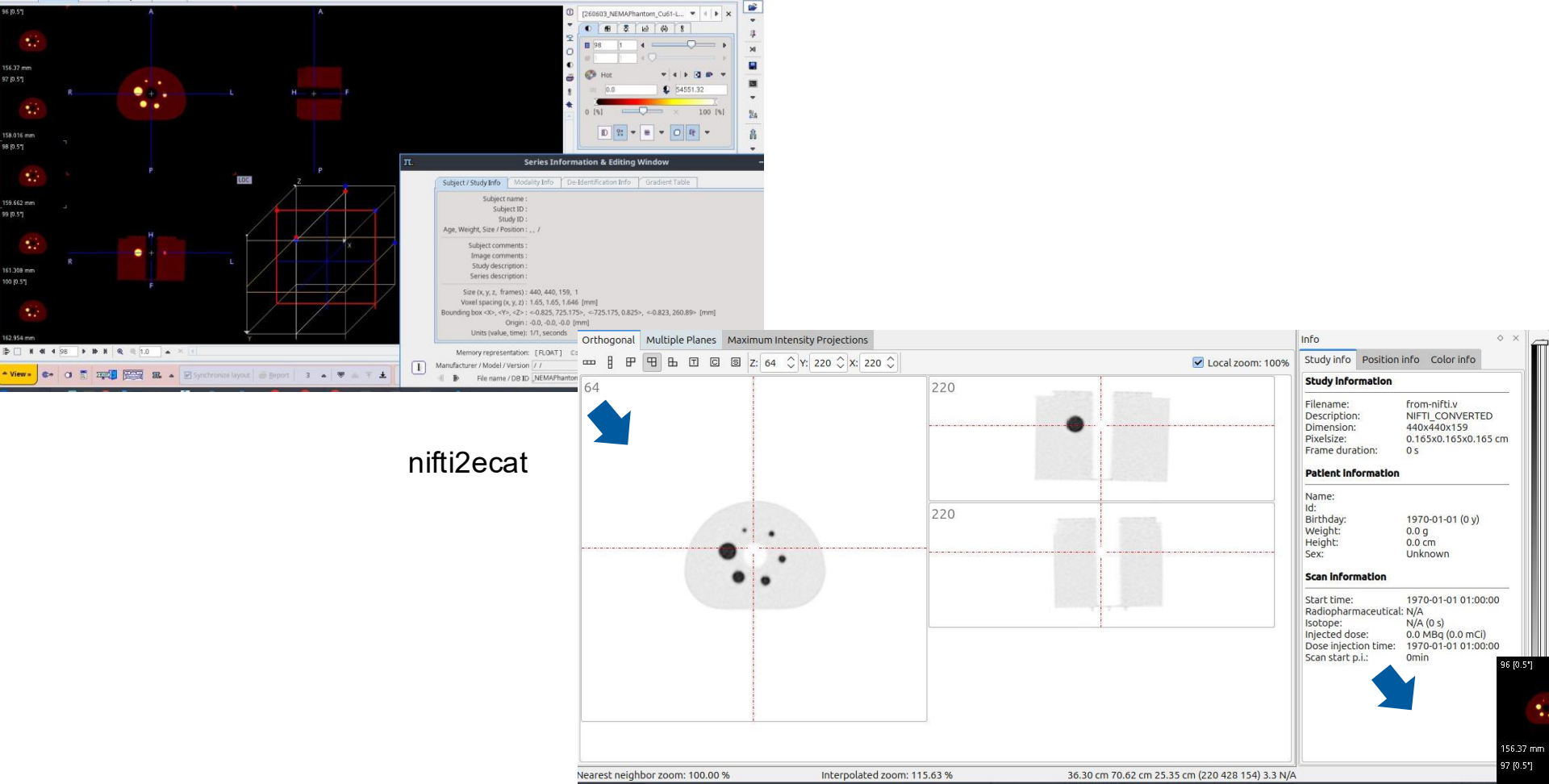
ecat2nifti



nifti2ecat



- Because of the embedded JSON Sidecar/Extension, the conversion successfully recovers the **Cu-61 isotope** and **injection timing**. This proves that no clinical metadata was lost during the transition through the NIfTI format.
- The anatomical orientation (L/R, A/P, H/F) perfectly matches the original phantom.
- Spatial geometry is preserved without artifacts, proving the Qform quaternion and Sform affine transformations function.
- Voxel dimensions are correctly interpreted from the new header.



OrthogonalMultiple PlanesMaximum Intensity Projections

59

A

B

230

221

59

A

B

230

221

ROI data

PrintExport DICOMExport ROI dataExport image

mean value (SUV (BW))

0.458

0.458

Mask_1.1

Mask_1.1

A

B

Select: Mean

OK

Info

Study infoPosition infoColor info

Study Information

Filename:from-ecat-and-back.v
Description:260603_NEMAPhantom_Cu61-LM-00-P
Dimension:440x440x159
Pixelsize:0.165x0.165x0.165 cm
Frame duration:1800 s

Patient Information

Name:Phantom_NEMA_Cu61_kontrast8
Id:03062026
Birthdate:1926-06-03 (100 y)
Weight:9.6 Kg
Height:100.0 cm
Sex:Unknown

Scan Information

Start time:2026-06-03 19:05:03
Radiopharmaceutical:Cu-61 Solution
Isotope:Cu-61 (12020.6 s)
Injected dose:60.8 MBq (1.6 mCi)
Dose injection time:2026-06-03 17:40:00
Scan start p.i.:1h, 25min, 3s

ROI Finder

Method: Fixed threshold

Run...

Identify ROIs

Apply to: All

In percent of max

Absolute

Lower: 0.0

Upper: None

Minimum volume (ccm): None

Enable segmentation

Exclude ROI intersection

Exclude Mask intersection

Shrink/grow ROI

File Tree

Name	ROI Volume [ccm]	ROI Max	ROI Mean	ROI Peak	ROI Mean Total	Asphericity
Project Phantom_NEMA_Cu61_kontrast8	-	-	-	-	-	-
Phantom_NEMA_Cu61_kontrast8 (PET)	-	-	-	-	-	-
Project Phantom_NEMA_Cu61_kontrast8	-	-	-	-	-	-
Phantom_NEMA_Cu61_kontrast8 (PET)	-	-	-	-	-	-
Project Phantom_NEMA_Cu61_kontrast8, ...	-	-	-	-	-	-
A: Phantom_NEMA_Cu61_kontrast8 (PET)	24086.8	11.5 SUV (BW)	0.5 SUV (BW)	7.9 SUV (BW)	11035.8 SUV (BW)*ccm	0.0
Mask_1	24086.8	11.5 SUV (BW)	0.5 SUV (BW)	7.9 SUV (BW)	11035.8 SUV (BW)*ccm	0.0
B: Phantom_NEMA_Cu61_kontrast8 (PET)	24086.8	11.5 SUV (BW)	0.5 SUV (BW)	7.9 SUV (BW)	11035.8 SUV (BW)*ccm	0.0
Mask_1	24086.8	11.5 SUV (BW)	0.5 SUV (BW)	7.9 SUV (BW)	11035.8 SUV (BW)*ccm	0.0
Project Phantom_NEMA_Cu61_kontrast8, ...	-	-	-	-	-	-
Phantom_NEMA_Cu61_kontrast8 (PET)	137316.5	-	-	-	-	0.0
Mask_1	137316.5	-	-	-	-	0.0
Mask_1.1	137316.5	-	-	-	-	0.0

Nearest neighbor zoom: 100.00 %

Interpolated zoom: 90.92 %

52.80 cm 8.41 cm 9.71 cm (320 51 59) A: 0.0 SUV (BW) B: 0.0 SUV (BW)

11

26/08/2026

Medical Imaging Data Formats

DRESDEN
concept

HZDR

```
giovan37@saturn:~/Documents/Libmedios$ ecaticmp -v /pet/projekte/libmedios/ecat/260603_NEMAPhantom_Cu61-LM-00-PSFTOF.v from-ecat-and-back.v
ecaticmp 1.12 - a libmedios ECAT6/7 file/data compare tool
(Apr 19 2024) Copyright (C) 2006-2024 Jens Maus, hzdr.de
```

```
Successfully loaded file1: '/pet/projekte/libmedios/ecat/260603_NEMAPhantom_Cu61-LM-00-PSFTOF.v'
Successfully loaded file2: 'from-ecat-and-back.v'
```

```
1: Checking META data:
  FileSize: 61566464:61566464 ...ok!
  Frames #: 1:1 ...ok!
  Planes #: 1:1 ...ok!
  Gates #: 1:1 ...ok!
  BedPos #: 0:0 ...ok!
```

```
2: Checking main header data:
ok!
```

```
Checking for header entry differences:
  CALIBRATION_UNITS.....: '1' != '0'
  ACQUISITION_TYPE.....: '3' != '4'
  LWR_TRUE_THRES.....: '0' != '350'
  UPR_TRUE_THRES.....: '0' != '650'
```

```
3: Checking directory list:
  Number of direntries: 1:1 ...ok!
```

#	Matrix-ID	SubHdrPos	MatrEndPos	Stat	Matrix-ID	SubHdrPos	MatrEndPos	Stat
01	01010001	0000001024	0061565952	1	01010001	0000001024	0061565952	1

```
4: Checking directory entries:
  DirItem #01: found in both files, size matches, status is equal... ok!
```

```
5: Checking for differences in subheaders:
  DirItem #01:
    RECON_ZOOM.....: '0' != '1'
    IMAGE_MAX.....: '32767' != '0'
    DECAY_CORR_FCTR.....: '1.05279' != '0'
    ANNOTATION.....: 'ROVER01' != 'Converted with libmedios'
    RECON_TYPE.....: '7' != '0'
    USER_RESERVED[00].....: '21071' != '0'
    USER_RESERVED[01].....: '22085' != '0'
    USER_RESERVED[02].....: '82' != '0'
    USER_RESERVED[06].....: '1' != '0'
    USER_RESERVED[11].....: '2026' != '0'
    USER_RESERVED[12].....: '6' != '0'
    USER_RESERVED[13].....: '3' != '0'
    USER_RESERVED[14].....: '19' != '0'
    USER_RESERVED[15].....: '4' != '0'
    USER_RESERVED[16].....: '56' != '0'
```

```
6: Checking datatype & RAW matrix data:
  Matrix: 1010001 Type: 6:6... ok!
  loading data... file1 ok, file2... ok!
  comparing data size 61564928:61564928... ok!
  comparing raw data... equal data... ok!
  comparing min/max values in matrix data:
    calculating (min:max) of data item #1... 0.000000e+00 : 3.276700e+04
    calculating (min:max) of data item #2... 0.000000e+00 : 3.276700e+04
    calculating max.count of data item #1... 6.998014e+09 ( 1.165047e+10) : 0.000000e+00 ( 0.000000e+00)
    calculating max.count of data item #2... 6.998014e+09 ( 1.165047e+10) : 0.000000e+00 ( 0.000000e+00)
```

```
Thanks for using libmedios!.. bye
```

```
giovan37@saturn:~/Documents/Libmedios$ ecaticmp -v /pet/projekte/libmedios/ecat/260603_NEMAPhantom_Cu61-LM-00-PSFTOF.v from-ecat-and-back.v
```

Outlook

What's next?

1. Implement Interfile format into libmedIO;
2. Spatial reorientation has to be implemented directly into the library → Universal interpretation of voxel data.

Thanks for your attention!



Summer Student program 2026 **Enhanced imaging data format support for AI in PET**

Author: Andrea Giovannelli

Supervisors: Jens Maus, Pavel Nikulin

DRESDEN
concept

